

Measurement

MCQ

1. Measurement is the comparison of an unknown quantity with a known fixed quantity called

A) Number
B) Unit
C) Measurement
D) Scale

Answer: B) Unit

2. The two parts of a measurement are

A) Length and breadth
B) Number and unit
C) Mass and time
D) Length and weight

Answer: B) Number and unit

3. Which of the following is a traditional unit of measurement?

A) Cubit
B) Metre
C) Kilometre
D) Litre

Answer: A) Cubit

4. The unit 'hand span' was used for measuring

A) Time
B) Length
C) Mass
D) Temperature

Answer: B) Length

5. The lack of uniformity in traditional measurements was because

- A) They were based on instruments
- B) They varied from person to person
- C) They used fixed standards
- D) They were very accurate

Answer: B) They varied from person to person

6. The SI system of units was developed in

- A) India
- B) France
- C) England
- D) America

Answer: B) France

7. The SI unit of length is

- A) Kilogram
- B) Litre
- C) Metre
- D) Centimetre

Answer: C) Metre

8. The SI unit of mass is

- A) Gram
- B) Kilogram
- C) Milligram
- D) Tonne

Answer: B) Kilogram

9. The SI unit of time is

- A) Hour
- B) Minute
- C) Second

D) Day

Answer: C) Second

10. For measuring very small lengths, the unit used is

A) Metre

B) Kilometre

C) Millimetre

D) Hectometre

Answer: C) Millimetre

11. The unit used to measure very long distances between cities is

A) Metre

B) Kilometre

C) Centimetre

D) Millimetre

Answer: B) Kilometre

12. 1 kilometre is equal to

A) 10 metres

B) 100 metres

C) 1000 metres

D) 10,000 metres

Answer: C) 1000 metres

13. 1 metre is equal to

A) 10 centimetres

B) 100 centimetres

C) 1000 centimetres

D) 10000 centimetres

Answer: B) 100 centimetres

14.1 centimetre is equal to

- A) 10 millimetres
- B) 100 millimetres
- C) 1000 millimetres
- D) 1 millimetre

Answer: A) 10 millimetres

15.The distance travelled by light in one year is called

- A) Light day
- B) Light speed
- C) Light year
- D) Solar year

Answer: C) Light year

16.The speed of light is approximately

- A) 3×10^5 km/s
- B) 3×10^6 km/s
- C) 3×10^4 km/s
- D) 3×10^3 km/s

Answer: A) 3×10^5 km/s

17.The unit used by sailors to measure distance at sea is

- A) Mile
- B) Nautical mile
- C) Kilometre
- D) Light year

Answer: B) Nautical mile

18.1 nautical mile equals

- A) 1 km
- B) 1.6 km
- C) 1.852 km

D) 2 km

Answer: C) 1.852 km

19. The measuring tool invented by William Bedwell is

A) Tape

B) Scale or ruler

C) Balance

D) Stopwatch

Answer: B) Scale or ruler

20. An odometer is used to measure

A) Mass

B) Volume

C) Distance travelled by vehicles

D) Time

Answer: C) Distance travelled by vehicles

21. The smallest division on a centimetre scale denotes

A) Metre

B) Centimetre

C) Millimetre

D) Inch

Answer: C) Millimetre

22. Parallax error occurs due to

A) Wrong instrument

B) Improper eye position

C) Faulty measurement unit

D) Temperature change

Answer: B) Improper eye position

23. The error caused when the object appears shifted due to different viewing angles is called

- A) Parallel error
- B) Parallax error
- C) Measurement error
- D) Visual error

Answer: B) Parallax error

24. The curved line can be measured using

- A) Ruler only
- B) Divider or thread
- C) Scale
- D) Compass

Answer: B) Divider or thread

25. 1 km = how many centimetres?

- A) 1000 cm
- B) 10,000 cm
- C) 100,000 cm
- D) 1,000,000 cm

Answer: C) 100,000 cm

26. The area of a rectangle is given by the formula

- A) $l + b$
- B) $l \times b$
- C) $l - b$
- D) $2(l + b)$

Answer: B) $l \times b$

27. If length = 20 m and breadth = 8 m, area = ?

- A) 28 m²
- B) 160 m²

C) 180 m^2

D) 200 m^2

Answer: B) 160 m^2

28. The SI unit of area is

A) m

B) m^2

C) m^3

D) cm^2

Answer: B) m^2

29. The formula for volume of a solid box is

A) $l + b + h$

B) $l \times b \times h$

C) $2(l + b + h)$

D) $l/b/h$

Answer: B) $l \times b \times h$

30. The SI unit of volume is

A) m^2

B) m^3

C) cm^2

D) cm^3

Answer: B) m^3

31. A cube with sides of 10 cm has a volume of

A) 100 cm^3

B) 1000 cm^3

C) 10 cm^3

D) 500 cm^3

Answer: B) 1000 cm^3

32.The unit used to measure volume of liquid is

- A) Kilogram
- B) Litre
- C) Metre
- D) Gram

Answer: B) Litre

33.1 litre is equal to

- A) 1000 ml
- B) 100 ml
- C) 10 ml
- D) 500 ml

Answer: A) 1000 ml

34.The instrument used to measure liquid volume accurately is

- A) Ruler
- B) Graduated cylinder
- C) Stopwatch
- D) Thermometer

Answer: B) Graduated cylinder

35.The water displacement method was discovered by

- A) Newton
- B) Einstein
- C) Archimedes
- D) Galileo

Answer: C) Archimedes

36.If water level rises from 150 ml to 180 ml, the volume of the object is

- A) 10 ml
- B) 20 ml
- C) 30 ml

D) 40 ml

Answer: C) 30 ml

37.1 ml equals

A) 1 cm³

B) 1 m³

C) 10 cm³

D) 100 cm³

Answer: A) 1 cm³

38. Gases can be measured in

A) Metres

B) Cubic metres or litres

C) Centimetres

D) Inches

Answer: B) Cubic metres or litres

39. 1 m³ =

A) 1 litre

B) 100 litres

C) 1000 litres

D) 10 litres

Answer: C) 1000 litres

40. The mass of an object is the measure of

A) Force

B) Quantity of matter

C) Weight

D) Pressure

Answer: B) Quantity of matter

41. The SI unit of mass is

- A) Gram
- B) Kilogram
- C) Milligram
- D) Tonne

Answer: B) Kilogram

42. Weight is the

- A) Volume of an object
- B) Gravitational force on mass
- C) Density
- D) Area

Answer: B) Gravitational force on mass

43. The Moon's gravity is how much of Earth's gravity?

- A) Half
- B) One-third
- C) One-sixth
- D) Equal

Answer: C) One-sixth

44. 1 kilogram =

- A) 10 grams
- B) 100 grams
- C) 1000 grams
- D) 10000 grams

Answer: C) 1000 grams

45. 1 tonne =

- A) 100 grams
- B) 100 kilograms
- C) 1000 kilograms

D) 10 kilograms

Answer: C) 1000 kilograms

46. The mass of a platinum-iridium bar kept in France represents

A) 1 gram

B) 1 kilogram

C) 1 milligram

D) 1 tonne

Answer: B) 1 kilogram

47. The beam balance measures

A) Length

B) Time

C) Mass

D) Area

Answer: C) Mass

48. The standard mass used in beam balance is called

A) Scale

B) Standard weight

C) Beam

D) Lever

Answer: B) Standard weight

49. Electronic balance is used for

A) Measuring time

B) Measuring volume

C) Measuring weight precisely

D) Measuring temperature

Answer: C) Measuring weight precisely

50. In laboratories, mass of chemicals is measured using

- A) Beam balance
- B) Electronic balance
- C) Thermometer
- D) Ruler

Answer: B) Electronic balance

51. The ancient instrument used to measure time using sand is

- A) Sundial
- B) Sand clock
- C) Water clock
- D) Pendulum

Answer: B) Sand clock

52. A sundial measures time based on

- A) Flow of sand
- B) Flow of water
- C) Shadow movement
- D) Pendulum swings

Answer: C) Shadow movement

53. A water clock measures time using

- A) Light
- B) Shadows
- C) Water flow
- D) Sound

Answer: C) Water flow

54. The most accurate device for measuring small durations is

- A) Sand clock
- B) Sundial
- C) Stopwatch

D) Candle clock

Answer: C) Stopwatch

55. The smallest unit of time is

A) Second

B) Minute

C) Hour

D) Day

Answer: A) Second

56. 1 hour =

A) 60 minutes

B) 100 minutes

C) 120 minutes

D) 30 minutes

Answer: A) 60 minutes

57. 1 minute =

A) 10 seconds

B) 30 seconds

C) 60 seconds

D) 100 seconds

Answer: C) 60 seconds

58. Ancient people measured time using

A) Thermometers

B) Clocks

C) Shadows

D) Batteries

Answer: C) Shadows

59. Inaccurate time measurement using shadows was due to

- A) Sunlight changes
- B) No clocks
- C) Incorrect tools
- D) Moonlight

Answer: A) Sunlight changes

60. The SI unit of temperature is

- A) Celsius
- B) Kelvin
- C) Fahrenheit
- D) Degree

Answer: B) Kelvin

61. Milli means

- A) 10^{-3}
- B) 10^{-2}
- C) 10^{-1}
- D) 10^3

Answer: A) 10^{-3}

62. Centi means

- A) 10^{-2}
- B) 10^{-3}
- C) 10^3
- D) 10^{-1}

Answer: A) 10^{-2}

63. Kilo means

- A) 10^3
- B) 10^{-3}
- C) 10^{-2}

D) 10^{-6}

Answer: A) 10^3

64. Nano means

A) 10^{-6}

B) 10^{-9}

C) 10^{-3}

D) 10^{-12}

Answer: B) 10^{-9}

65. Deci means

A) 10^{-3}

B) 10^{-2}

C) 10^{-1}

D) 10^3

Answer: C) 10^{-1}

66. The measuring tape is used to measure

A) Time

B) Long distances

C) Volume

D) Temperature

Answer: B) Long distances

67. The device that measures distance in vehicles is

A) Speedometer

B) Odometer

C) Tachometer

D) Ruler

Answer: B) Odometer

68. The beam balance compares

- A) Force
- B) Two masses
- C) Two volumes
- D) Two lengths

Answer: B) Two masses

69. The device used to measure very short time intervals is

- A) Clock
- B) Stopwatch
- C) Calendar
- D) Timer

Answer: B) Stopwatch

70. 1 litre = how many cubic centimetres?

- A) 10 cm^3
- B) 100 cm^3
- C) 1000 cm^3
- D) $10,000 \text{ cm}^3$

Answer: C) 1000 cm^3

71. Volume of a cube =

- A) Side $\times$ Side
- B) Side³
- C) 2 $\times$ Side
- D) Side²

Answer: B) Side³

72. Measuring jar is used for

- A) Measuring mass
- B) Measuring volume of liquid
- C) Measuring distance

D) Measuring time

Answer: B) Measuring volume of liquid

73. A pipette is used to measure

A) Air volume

B) Small liquid quantities

C) Weight

D) Height

Answer: B) Small liquid quantities

74. The divider is used to measure

A) Curved lines

B) Straight lines

C) Mass

D) Time

Answer: A) Curved lines

75. Volume of irregular object is found by

A) Measuring tape

B) Water displacement

C) Ruler

D) Scale

Answer: B) Water displacement

76. The relation between 1 ml and 1 cm³ is

A) 1 ml = 1 cm³

B) 1 ml = 10 cm³

C) 1 ml = 100 cm³

D) 1 ml = 0.1 cm³

Answer: A) 1 ml = 1 cm³

77. 1 kg = how many milligrams?

- A) 100 mg
- B) 1000 mg
- C) 1,000,000 mg
- D) 10,000 mg

Answer: C) 1,000,000 mg

78. The mass of 1 litre of water is approximately

- A) 1 g
- B) 10 g
- C) 100 g
- D) 1 kg

Answer: D) 1 kg

79. Area depends on

- A) One dimension
- B) Two dimensions
- C) Three dimensions
- D) No dimension

Answer: B) Two dimensions

80. Volume depends on

- A) One dimension
- B) Two dimensions
- C) Three dimensions
- D) No dimension

Answer: C) Three dimensions

81. A metre rod is used for measuring

- A) Area
- B) Length
- C) Volume

D) Time

Answer: B) Length

82. Measuring instrument used in jewellery shops is

A) Beam balance

B) Electronic balance

C) Ruler

D) Graduated cylinder

Answer: B) Electronic balance

83. The basic unit of length in SI is

A) Centimetre

B) Metre

C) Millimetre

D) Kilometre

Answer: B) Metre

84. One astronomical unit measures distance between

A) Moon and Earth

B) Earth and Sun

C) Planets and Stars

D) Two stars

Answer: B) Earth and Sun

85. The total space occupied by an object is called its

A) Area

B) Mass

C) Volume

D) Height

Answer: C) Volume

86. A thread can be used to measure

- A) Straight line
- B) Curved line
- C) Angle
- D) Weight

Answer: B) Curved line

87. An example of a curved object is

- A) Book
- B) Plate
- C) Circle
- D) Ruler

Answer: C) Circle

88. The time measuring device in mobile phones is

- A) Calendar
- B) Stopwatch
- C) Clock
- D) Timer

Answer: B) Stopwatch

89. 1 centimetre = how many metres?

- A) 0.001 m
- B) 0.01 m
- C) 0.1 m
- D) 0.0001 m

Answer: B) 0.01 m

90. A solid with equal length, breadth and height is called

- A) Rectangle
- B) Square
- C) Cube

D) Cuboid

Answer: C) Cube

91. Volume of cube with side 5 cm =

A) 25 cm^3

B) 50 cm^3

C) 125 cm^3

D) 225 cm^3

Answer: C) 125 cm^3

92. The Earth's gravity pulls all objects

A) Downward

B) Upward

C) Sideways

D) Away

Answer: A) Downward

93. On the Moon, the weight of an object becomes

A) Six times more

B) Six times less

C) Same

D) Double

Answer: B) Six times less

94. The measuring unit used for fuel quantity is

A) Gram

B) Litre

C) Metre

D) Kilogram

Answer: B) Litre

95.The SI unit for area is derived from

- A) One length
- B) Two lengths
- C) Three lengths
- D) Volume

Answer: B) Two lengths

96.Which among the following can be compressed easily?

- A) Solids
- B) Liquids
- C) Gases
- D) Metals

Answer: C) Gases

97.The standard unit for time used worldwide is

- A) Second
- B) Minute
- C) Hour
- D) Day

Answer: A) Second

98.Measurement helps us to

- A) Guess distances
- B) Compare quantities accurately
- C) Reduce errors fully
- D) Play games

Answer: B) Compare quantities accurately

99.The standard unit used to measure liquid quantities like milk or water is

—

- A) Metre
- B) Kilogram

C) Litre

D) Centimetre

Answer: Option C) Litre

100. Which of the following statements about measurement is true?

A) Measurement does not need standard units

B) Standard units help in uniform and accurate measurement everywhere

C) Different places can have different standard units for the same quantity

D) Measurement can only be done using traditional methods

Answer: Option B) Standard units help in uniform and accurate measurement everywhere